CSE 4711 (CSE): AI - Quiz #2 **16 October 2024**

Duration: 30 Minutes **Full Marks: 15**

1. For each of the following scenarios, formulate it as a Constraint Satisfaction Problem (CSP) by identifying the variables, their possible values, and the constraints (You can provide your answer in plain English): 2×6 (CO2) (PO2)

   a) The knight is a chess piece symbolized by the head and neck of a horse. It moves in an "L" shape: either two squares vertically and one square horizontally, or two squares horizontally and one square vertically. It can leap over other pieces on the board.

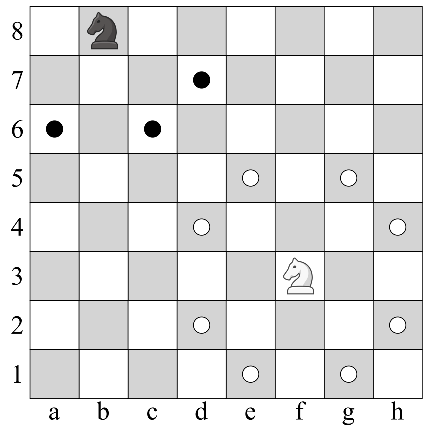

Figure 1: Knights on different squares of an 8×8 chessboard, with dots representing possible knight moves

You have to place k knights on an $n \times n$ chessboard such that no two knights are attacking each other, where $k \leq n^2$.

Solution:

Approach 1:

- **Variables:** X_{rc}, where r indicates the row and c indicates the column in the chessboard. So one variable for each cell.
- **Values:** Each variable can take one of two values, $\{0, 1\}$ indicating whether the corresponding cell has a knight or not.
- **Constraints:**
  - Every pair of squares separated by a knight's move is constrained, such that both cannot be occupied.
  - The entire set of squares is constrained, such that the total number of occupied squares should be k.

Approach 2:

- **Variables:** X_k, where k indicates the kth knight. So one variable for each knight.
- **Values:** Each variable's can take the values (r, c) where $1 \leq r, c \leq n$.
- **Constraints:** Every pair of variables is constrained, such that no two knights can be on the same square or on squares separated by a knight's move.

Rubric:

- 2 points for variable with appropriate description
- 2 points for values with appropriate description
- 2 points for constraint(s)

   b) A crossword puzzle is a word game featuring an $m \times n$ grid made up of shaded and unshaded squares. The shaded squares stay blank, while the unshaded squares are filled with words that intersect both horizontally and vertically.

   Assume that a list of English words (i.e., a dictionary) is provided. You have to construct a crossword puzzle by filling up the empty squares using a subset of the dictionary.

	A	C	E	
G	L	O	V	E
A	P	R	I	L
S	H	A	C	K
	A	L	T	

Figure 2: A solved 5×5 crossword puzzle

Solution:

Approach 1:

- **Variables:** Each empty box can be considered as a variable
- **Values:** Each letter from the English alphabet.
- **Constraints:** Each string of consecutive horizontal or vertical boxes must make a word from the dictionary.

Approach 2:

- **Variables:** Each string of consecutive horizontal or vertical boxes can be a single variable.
- **Domain:** Words from the dictionary of the right length.
- **Constraints:** Two intersecting words must have the same letter in the intersecting box.

Rubric:

- 2 points for variable with appropriate description
- 2 points for values with appropriate description
- 2 points for constraint(s)

2. Suppose we are solving a CSP using local search. The problem has N variables, each having a domain with d values. We want to generate successors of any state: from a given complete assignment to all variables, what other assignments we will consider. For each of the following approaches of generating successors, state the worst-case computational complexity in terms of N and d: 1×3 (CO1) (PO1)

   a) Randomly choose a variable, then consider all assignments to that variable.

Solution:

$O(d)$ (since we have to go through all the values of that variable)

Rubric:

- 1 point for the exact complexity
- (You do not have to write the explanation, as it was not asked. There are provided here for your understanding purpose.)

   b) For the variable with the most conflicts, consider all assignments to it.

Solution:

$O(d + N)$ (since we might have to iterate through all the variables to determine the one with the most conflict)

Rubric:

- 1 point for the exact complexity
- (You do not have to write the explanation, as it was not asked. There are provided here for your understanding purpose.)

   c) Consider all states that differ by the assignment to exactly one variable.

Solution:

$O(Nd)$ (For each variable, consider each possible assignment)

Rubric:

- 1 point for the exact complexity
- (You do not have to write the explanation, as it was not asked. There are provided here for your understanding purpose.)